1641357

CSCHE202

VTU.REG.NO

B.E. Second Semester Examination – June/July 2026

Applied Chemistry for Smart System

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Module – I

1. a) Explain the following characteristics of battery: i) Capacity ii) Cycle life iii) Power density (06 Marks)
   b) Explain the construction, working and applications of lithium-ion battery. (08 Marks)
   c) Explain construction, working and application of Na-ion battery. (06 Marks)

2. a) Define secondary battery. Explain construction, working and applications of lead-acid secondary storage battery. (08 Marks)
   b) Explain construction, working and application of Ni-MH battery. (06 Marks)
   c) Define fuel cell. Explain construction, working and applications of solid oxide fuel cell (SOFCs). (06 Marks)

Module – II

3. a) Explain free radical mechanism of addition polymerization taking ethylene as an example. (06 Marks)
   b) In a polymer sample, 100 molecules have molecular weight of 10^3 g/mol 250 molecules have molecular weight of 10^4 g/mol and 300 molecules have molecular weight of 10^5 g/mol, calculate $\overline{M}_n$, $\overline{M}_W$ (06 Marks)
   c) Define glass transition temperature. Explain the factors affecting glass transition temperature and its significance. (08 Marks)

4. a) Explain synthesis, properties and applications of PVC and PMMA. (06 Marks)
   b) Define conducting polymers. Explain synthesis and mechanism of conducting polyaniline molecule. (08 Marks)
   c) Explain synthesis, properties and industrial applications of cd-se, quantum dot by wet chemical method. (06 Marks)

Module – III

5. a) Define viscosity. Explain the determination of viscosity of the given liquid by using Ostwald's viscometer. (08 Marks)
   b) What are optical sensors? Explain the determination of amount of copper present in the given solution by using calorimetric sensor. (06 Marks)
   c) Explain the amount of iron present in the given solution of steel industry by using potentiometric sensors. (06 Marks)

6. a) Define corrosion. Explain electrochemical theory of corrosion taking iron as an example. (06 Marks)
   b) Define differential aeration corrosion. Explain waterline and pitting corrosion. (05 Marks)
   c) Explain the following factor affecting rate of corrosion : i) Nature of metal ii) Temperature iii) pH (09 Marks)

Module – IV

7. a) Explain production of green hydrogen by photocatalytical method using TiO_2. (06 Marks)
   b) Explain synthesis of Bio-Diesel by trans-esterification method. Mention its advantages and its applications. (06 Marks)
   c) Explain production of green-hydrogen by photoelectrocatalytic water splitting method mention its advantages. (08 Marks)

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8. a) Explain sources and composition of E-waste. (06 Marks)
   b) Explain use of artificial intelligence in E-waste management and its applications. (06 Marks)
   c) Explain extraction of gold from E-waste by using bioleaching method. (08 Marks)

Module – V

9. a) Explain sources and nature of impurities in water. (08 Marks)
   b) Calculate temporary, permanent and total hardness of sample of water, which contains 16.2 mg of $Ca(HCO_3)_2$, 29.2 mg of $Mg(HCO_3)_2$, 33.3 mg $CaCl_2$ and 18 mg of $MgSO_4$ (At.wt Ca = 40, Mg = 240 mu) per liter of water. (06 Marks)
   c) Explain Lime-soda process of softening of water. (06 Marks)

10. a) Illustrate purification of water by reverse osmosis and chlorination method. (08 Marks)
    b) Define Nano-particles and explain properties and engineering application of CNT. (06 Marks)
    c) Explain properties and engineering application of grapheme. (06 Marks)

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PDA COLLEGE OF ENGINEERING, KALABURAGI

(An autonomous institution, affiliated to VTU, Belagavi, Karnataka)

Dept. of Chemistry

USN | | | | | | | | | | |

Internal Assessment Test – II

Applied Chemistry for Smart Systems CSCHE202

Date: 25-05-2026 Time: 11.00am-12.15pm Maximum Marks: 40

Instructions to Candidates:
1. Answer all the following questions.

Q.NO	Question	Maximum Marks	CO	BL
1a.	State Beer-Lamberts law. How you estimate the amount of copper present in given solution by calorimetrically.	08	CO3	L2
b.	Illustrate Electrochemical theory of corrosion by taking iron as an example.	08	CO3	L3
c.	Explain how Temperature & pH affect rate of corrosion.	04	CO3	L2
2a.	Discuss the sources & composition of e–waste.	08	CO4	L2
b.	Outline the extraction of gold from e-waste by bioleaching method.	08	CO4	L3
c.	Explain the role of AI in e-waste management.	04	CO4	L2

Course Outcomes	Total Marks allotted	% Marks Allotted	Cognitive levels	Total Marks allotted	% Marks Allotted
CO3	20	50	L1-L2	24	60
CO4	20	50	L3	16	40

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Internal Assessment Test –II

22CHEE12 Chemistry for EEE Stream

Date: 23/12/2024 Time: 09.15am-10.30am Maximum Marks: 40

Instructions to Candidates:

Answer all the following questions.

Q.NO	Question	Maximum Marks	CO	BL
1a.	Define corrosion. Illustrate the electrochemical theory of corrosion by taking iron as an example.	08	CO3	L3
b.	Explain waterline corrosion.	04	CO3	L2
c.	Discuss corrosion control by galvanization and sacrificial anode method.	08	CO3	L2
2a.	Illustrate sources and nature of impurities present in water.	08	CO4	L3
b.	A sample of water on analysis was to contain $Ca(HCO_3)_2$-16.2mg, $MgSO_4$-3.6mg and $CaSO_4$-6.8mg per litre. Calculate temporary, permanent and total hardness. (at.wts. Ca-40, Mg-24, O-16, C-12, S-32, H-1)	08	CO4	L3
c.	Explain types of hardness.	04	CO4	L2

Course Outcomes	Total Marks allotted	% Marks Allotted	Cognitive levels	Total Marks allotted	% Marks Allotted
CO3	20	50	L1-L2	16	40
CO4	20	50	L4	24	60

Internal Assessment Test –III

22CHES22 Applied Chemistry for CSE Stream

Date: 03/07/2024 Time: 7.45am-9.00am Maximum Marks: 30

Instructions to Candidates:

Answer all the following questions.

Q.NO	Question	Maximum Marks	CO	BL
1a.	What are conducting polymer? Explain mechanism of counducting polymer of polyactylene.	06	CO4	L2
b.	Define Fibers? Explain synthesis, properties and industrial applications of Kevlar and polyester.	06	CO4	L2
c.	Illustrate chlorination method of purification of water.	03	CO5	L3
2a.	Illustrate sources and nature of impurities of water.	06	CO5	L3
b.	Calculate the temporary hardness, permanent hardness and total hardness of sample of water which contain 16.2 mg of $Ca(HCO_3)_2$, 29.2 mg of $Mg(HCO_3)_2$, 33.3 mg of $CaCl_2$, 18 mg of $MgSO_4$ and 55 mg of NaCl per litre of water.	06	CO5	L4
c.	Illustrate properties and engineering applications of graphene.	03	CO5	L3

Internal Assessment Test –I

DEPARTMENT OF CHEMISTRY

Course : Applied Chemistry for EC & EEE Stream **Course code:** 22CHEE12

Date: 11/11/2024 Time: 09.15 to 10:30 AM Maximum Marks: 40

Instructions to Candidates:

1. Answer all the following questions

Q.NO	Question	Maximum Marks	CO	BL
1a.	What are polymers? Explain free radical mechanism of addition polymerization of ethylene.	08	CO1	L3
b.	Define conducting polymers. Explain mechanism of conducting polyactylene.	08	CO1	L3
c.	Illustrate the preparation, properties and applications of Kevlar.	04	CO1	L2
2a.	Illustrate with neat labeled diagram, construction, working and applications of Zn-Air modern battery +	08	CO2	L2
b.	Define Secondary battery. Illustrate Construction, Working and Application of Lithium Ion Battery.	08	CO2	L2
c	Illustrate construction, working and application of Ni-MH modern battery.	04	CO2	L3

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Internal Assessment Test –I

DEPARTMENT OF CHEMISTRY

Course : Applied Chemistry for CSE Stream **Course code:** **22CHES22**

Date: 15/04/2024 Time: 03.00:04:15 PM Maximum Marks: 30

Instructions to Candidates:

1. Answer all the following questions

Q.NO	Question	Maximum Marks	CO	BL
1a.	Discuss the principle, working of conductometric titration of strong acid with strong base.	06	CO1	L2
b.	Illustrate the construction, working and applications of Li-ion battery.	06	CO1	L4
c.	Define fuel cell. Illustrate the construction, working and applications of CH_3OH-O_2 fuel cell.	06	CO1	L4
2a.	What is E-waste management? Give the sources and composition of E-waste.	06	CO2	L2
b.	Discuss the toxic materials used in manufacturing electronic and electrical products.	06	CO2	L2

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22CHEE12

VTU.REG.NO

B. E. First/Second Semester Examination – January 2025

Chemistry for EEE Stream

(Common to EEE, EI and ECE)

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE question from each module.

MODULE - I

1. a) Illustrate the free radical mechanism of addition polymerization taking ethylene as on example. **(06 Marks)**
   b) Explain the determination of number average and weight average molecular weight of polymer. **(08 Marks)**
   c) A polymer sample contains 1,2,3 and 4 molecules having molecular weight 10^5, 2×10^5 , 3×10^5 and 4×10^5 respectively. Calculate the number average and weight average molecular weight of polymer. **(06 Marks)**

2. a) Illustrate the synthesis , properties and industrial application of Kevlar. **(06 Marks)**
   b) Explain preparation, properties and applications of Teflon and Polymethyl methacrylate (PMMA). **(06 Marks)**
   c) What are conducting polymers? Illustrate the synthesis and conducting mechanism of polyacetylene. **(08 Marks)**

MODULE - II

3. a) Illustrate the construction, working and applications of Zn –Air modern battery. **(06 Marks)**
   b) *Discuss* the construction, working and application of Na-ion battery. **(07 Marks)**
   c) *Illustrate* the construction, working and applications of Ni-MH (Metal hydride) battery. **(07 Marks)**

4. a) What are fuel cells? Explain construction, working and application of methnol-oxygen fuel cell. **(06 Marks)**
   b) Define calorific value. Illustrate the determination of calorific value of solid fuel using Bomb calorimeter. **(08 Marks)**
   c) A 0.90 gm of fuel sample was completely combusted in excess of oxygen the increase of temperature of water on a calorimeter containing 1050 gm of water was 2.5°C. Calculate the grow calorific value of the fuel if water equivalent of calorimeter is 135 gm. Given specific heat of water is 4.187 kJ/kg. **(06 Marks)**

MODULE - III

5. a) Define corrosion? Illustrate the electrochemical theory of corrosion taking iron as an example. **(07 Marks)**
   b) Explain the galvanization methods of corrosion control. **(04 Marks)**
   c) Illustrate the following factors effecting corrosion.
   i) Temperature ii) pH iii) Relative area of anode and cathode. **(09 Marks)**

6. a) What is E-waste? Explain the types and sources of E-waste. **(07 Marks)**
   b) Illustrate the effect of E-waste on human health. **(06 Marks)**
   c) Illustrate the method of extraction of gold from E-waste. **(07 Marks)**

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MODULE - IV

7. a) Explain sources and nature of impurities present in water. **(08 Marks)**
   b) Explain the determination of total hardness of water by EDTA method. **(06 Marks)**
   c) A water sample contains $Ca(HCO_3)_2$ = 22.4 mg/L, $Mg(HCO_3)_2$ = 19.2 mg/L and $CaSO_4$ = 23.5mg/L. Calculate temporary, permanent and total hardness. Atomic weight of Ca = 40, O = 16, H = 1, C= 12, S = 32, Mg = 24. **(06 Marks)**

8. a) Define COD. Illustrate the determination COD of waste water.. **(08 Marks)**
   b) Explain the method of reverse Osmosis of water purification. **(06 Marks)**
   c) What are nanomaterials? Illustrate the properties of carbon nanotubes. **(06 Marks)**

MODULE - V

9. a) What are reference electrodes? Illustrate the construction and working of Calomel electrode. **(08 Marks)**
   b) What are concentration cells? Discuss the EMF of concentration cells. **(06 Marks)**
   c) A concentration cell is constructed by dipping copper rods in 0.001 m and 0.1 m $CuSO_4$ solution. Calculate EMF of the cell at 298°K. Write cell representation and cell reaction. **(06 Marks)**

10. a) Illustrate the method of estimation of amount of copper present in given solution using calorimeter. **(07 Marks)**
    b) Illustrate the conductometric filtration of strong and with strong base. **(07 Marks)**
    c) Explain the determination of amount of iron by potentiometric method. **(06 Marks)**

CHE12

22CHE/S22

VTU.REG.NO

B.E. Second Semester Examination – July 2024

Chemistry for Computer Science Stream

Time: 3 hrs] [Maximum Marks: 100

Note: Answer any FIVE full questions, selecting atleast ONE full question from each module.

Module – I

1. a) What is conductometric sensor? Interpret conductometric titration of strong acid with strong base. (07 Marks)
   b) Define battery. Explain classification of batteries with examples. (06 Marks)
   c) What is potentiometric type of electrochemical sensor? Outline about estimation of Fe by potentiometrically. (07 Marks)

2. a) Develop construction, working and applications of Na-ion battery. (07 Marks)
   b) Define Li-ion battery. Find the construction and working of Li-ion battery. (06 Marks)
   c) What are fuel cells? Organize construction, working and applications of CH_3OH-O_2 fuel cell. (07 Marks)

Module – II

3. a) What is e-waste? List the sources and composition of e-waste. (06 Marks)
   b) Interpret the toxic materials used in electronic and electrical products and their health hazards. (07 Marks)
   c) Discuss the characteristics and need of a e-waste management. (07 Marks)

4. a) Summarize separation and thermal treatment methods for recycling and recovery from e-waste. (06 Marks)
   b) Illustrate hydrometallurgical and pyrometallugical methods of extraction from e-waste. (07 Marks)
   c) What is the role of stake holders in environmental management of e-waste. (07 Marks)

Module – III

5. a) Define corrosion. Illustrate electrochemical theory of corrosion by taking iron as an example. (06 Marks)
   b) What is anodic coating? Explain control of corrosion by galvanization. (05 Marks)
   c) Outline about the following factors affecting the role of corrosion:
      i) Relative area of anode and cathode area ii) Temperature iii) pH (09 Marks)

6. a) What is concentration cell? A concentration cell is constructed by dipping copper rods in 0.001M and 0.1M $CuSO_4$ solutions. Calculate the emf of the cell at 298 K. (06 Marks)
   b) What are reference electrode? Summarize the construction, working of calomel electrode. (07 Marks)
   c) Illustrate the determination of pH using glass electrode. (07 Marks)

Module – IV

7. a) Define polymers. Outline free radical mechanism of addition polymerization taking ethylene as an example. (06 Marks)
   b) Organize synthesis, properties and applications polyvinylchloride (PVC). (06 Marks)
   c) What are conducting polymers? Explain conducting mechanism of polyacetylene. (08 Marks)

8. a) A polymer sample contains 1,2,3 and 4 molecules having molecular weights 10^5, 2×10^5, 3×10^5, 4×10^5 respectively. Calculate the number average and weight average molecular weight of the polymer. (06 Marks)
   b) What are fibers? Illustrate synthesis properties and applications of Kevlar. (07 Marks)
   c) What are plastics? Explain synthesis properties and applications of Teflon. (07 Marks)

Module – V

9. a) Illustrate about sources and nature of impurities of water. **(08 Marks)**
   b) Explain purification water by reverse osmosis with neat diagram. **(06 Marks)**
   c) Illustrate about the following: **(06 Marks)**
   i) Temporary hardness ii) Permanent hardness iii) Total hardness

10. a) Define COD. 20 ml of sewage sample for COD is reacted with 25 ml of $K_2Cr_2O_7$ solution and the unreacted $K_2Cr_2O_7$ requires 9.0 ml of N/4 FAS solution. Under similar conditions, in blank titration 15 ml of FAS is used up. Calculate the COD of the sample. **(07 Marks)**
    b) Organize the properties and engineering applications of carbon nanotubes. **(06 Marks)**
    c) Explain properties and engineering applications of graphene. **(07 Marks)**